

Long-term values in Dynamic Programming and Stochastic Games

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Abstract

The course will first consider discrete-time dynamic programming problems and Markov decision processes, focusing on the limit value for long time horizon. We will then consider the extension to zero-sum stochastic games, where the state variable is controlled by two players with opposite interests. Time permitting, we will also discuss the existence of the uniform value, where players are required to play epsilon-optimally simultaneously in all problems with sufficiently long time horizon.